Fracture Mechanics for Delamination Problems in Composite Materials

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ABSTRACT

A fracture mechanics approach to the well-known delamination problem in composite materials is presented. Based on the theory of anisotropic laminate elasticity and interlaminar fracture mechanics concepts, the composite delamination problem is formulated and solved. The exact order of the delamination crack-tip stress singularity is determined. Asymptotic stress and displacement fields for an interlaminar crack are obtained. Fracture mechanics parameters such as mixed-mode stress intensity factors, K_I, K_{II}, K_{III}, and the energy release rate, G, for composite delamination problems are defined. To illustrate the fundamental nature of the delamination crack behavior, solutions for edge-delaminated graphite-epoxy composites under uniform axial extension are presented. Effects of fiber orientation, ply thickness, and delamination length on the interlaminar fracture are examined.

1. INTRODUCTION

DELAMINATION, SOMETIMES ALSO CALLED INTERLAMINAR CRACKING, IS one of the most frequently encountered types of damage in advanced composite materials. The problem is not only of great theoretical interest but also of significant technical importance. The presence and growth of delamination cracks in composite laminates may lead to severe reliability and safety problems, such as reduction of structural stiffness, exposure of the interior to adverse environment, and disintegration of the material, which may cause the final fracture. Thus, understanding the basic mechanics of delamination is of critical importance in characterization of flaw criticality and assessment of structural integrity of advanced composite materials and structures.

The delamination problem is generally very complex in nature and difficult to solve, because it involves not only geometric and material discontinuities,

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but also the inherently coupled mode I, II and III fracture in an anisotropic, layered material system. This situation is especially true for angle-ply composite laminates, in which delamination is inherently three-dimensional in nature. The composite delamination is basically an interlaminar fracture problem involving debonding or separation between two highly anisotropic, fiber-reinforced laminae. The problem of an interfacial crack between two dissimilar isotropic materials has received much attention recently, for example, References 1–12. However, systematic studies on the subject of delamination between dissimilar, anisotropic fiber composites have been very limited because of the complexities involved.

In this paper, a study on the fundamental nature of the delamination problem based on recently developed interlaminar fracture mechanics theory for composite materials [13–15] is presented. Governing partial differential equations derived from the theory of anisotropic elasticity and general solutions for the composite mechanics problem are briefly outlined. Stress singularities for a delamination crack in the composite laminate fracture are determined. Fracture mechanics parameters such as mixed-mode stress intensity factors K_I, K_{II} and K_{III} as well as the energy release rate G are introduced. To illustrate the fundamental mechanics of delamination, numerical examples on the important and well-known problem of edge-delaminated composites subjected to uniform axial extension are presented. Effects of geometric, lamination, and crack variables on the interlaminar fracture behavior are studied also.

2. ASSUMPTIONS

Consider a composite material (Fig. 1) composed of unidirectional fiber-reinforced plies of uniform thicknesses, h_1, h_2 . . ., h_n. For simplicity and

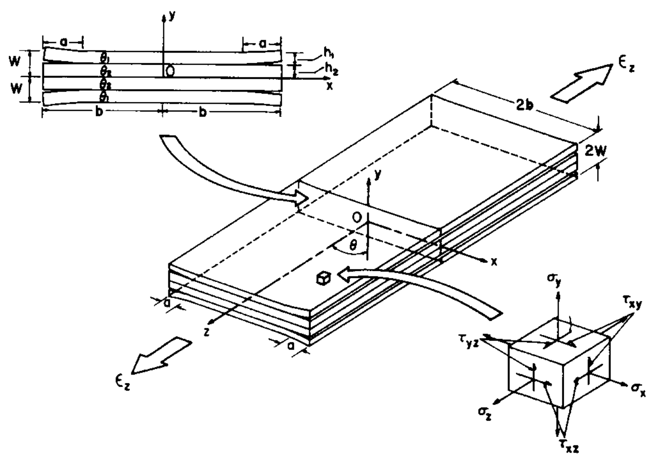

Figure 1. Delamination cracks in a $[\theta_1/\theta_2/\theta_2/\theta_1]$ composite laminate under uniform axial extension $\varepsilon_z = e$.

without loss of generality, we restrict our attention to the cases of symmetric composite laminates with $[\theta_1/\theta_2/\ldots/\theta_2/\theta_1]$ fiber orientations and ply thicknesses, $h_1 = h_n$, $h_2 = h_{n-1}$, ... The composite has a width $2b$ and is subjected to a uniform axial extension, $\varepsilon_z = e$, along the z axis. The composite laminate is sufficiently long that, in the region far away from the ends, end effects are negligible by virtue of Saint Venant's principle. Consequently, stresses in the composite are independent of the z axis. Delamination occurs in the form of interlaminar cracks between dissimilar plies with fiber orientations θ_k and θ_{k+1}. Perfect bonding is assumed in the composite everywhere except the region of delamination.

3. BASIC EQUATIONS AND GENERAL SOLUTIONS

A theory for the mechanics of delamination fracture in composite materials has been developed recently by the author, and some of the preliminary information have been reported in [13,14]. Due to space limitation, only a brief outline of the theory is given here; detailed mathematical derivation, solution accuracy and convergence, and interlaminar fracture mechanics interpretations are given in [13–15]. It is only stated that in this paper formulation of the delamination problem is based on the theory of anisotropic laminate elasticity and that well-known Lekhnitskii's stress functions [16] are used to establish governing partial differential equations. An eigenfunction expansion method is employed for determining the stress singularity at the delamination crack tip. The boundary collocation technique is then used to evaluate the complete solution for finite-dimensional composite laminates having delamination cracks under mechanical loading.

3.1 Governing Partial Differential Equations

For each individual fiber-reinforced lamina with rectilinear anisotropy of a general form, the constitutive equations may be expressed by generalized Hooke's law in contracted notation as

$$\varepsilon_i = S_{ij}\,\sigma_j \qquad (i,j = 1,2,\ldots,6), \tag{1}$$

where ε_i and σ_j are strain and stress components, and S_{ij} is the compliance tensor, respectively. The engineering strains, ε_i, are defined in a Cartesian coordinate system by $\varepsilon_1 = \varepsilon_x$, $\varepsilon_2 = \varepsilon_y$, . . . , $\varepsilon_5 = 2\varepsilon_{xz}$, $\varepsilon_6 = 2\varepsilon_{xy}$, and the stresses σ_i are defined in an analogous manner.

Introducing Lekhnitskii's stress potentials [16], $F(x,y)$ and $\Psi(x,y)$, and following the procedure given in [13,16], one can obtain a pair of coupled, governing partial differential equations for each composite lamina as

$$\begin{cases} L_4F + L_3\Psi = 0, & \text{(2a)} \\ L_3F + L_2\Psi = 0, & \text{(2b)} \end{cases}$$

where L_2, L_3 and L_4 are linear differential operators of the second, third and fourth order, respectively, defined by

$$L_2 = \tilde{S}_{44} \frac{\partial^2}{\partial x^2} - 2\tilde{S}_{45} \frac{\partial^2}{\partial x \partial y} + \tilde{S}_{55} \frac{\partial^2}{\partial y^2}, \tag{3a}$$

$$L_3 = -\tilde{S}_{24} \frac{\partial^3}{\partial x^3} + (\tilde{S}_{25} + \tilde{S}_{46}) \frac{\partial^3}{\partial x^2 \partial y} - (\tilde{S}_{14} + \tilde{S}_{56}) \frac{\partial^3}{\partial x \partial y^2} + \tilde{S}_{15} \frac{\partial^3}{\partial y^3}, \tag{3b}$$

$$L_4 = \tilde{S}_{22} \frac{\partial^4}{\partial x^4} - 2\tilde{S}_{26} \frac{\partial^4}{\partial x^3 \partial y} + (2\tilde{S}_{12} + \tilde{S}_{66}) \frac{\partial^4}{\partial x^2 \partial y^2} - 2\tilde{S}_{16} \frac{\partial^4}{\partial x \partial y^3} + \tilde{S}_{11} \frac{\partial^4}{\partial y^4}, \tag{3c}$$

in which $\tilde{S}_{ij}$ is the reduced compliance tensor defined as

$$\tilde{S}_{ij} = S_{ij} - S_{3i} S_{3j}/S_{33} \qquad (i,j = 1,2,4,5,6). \tag{3d}$$

3.2 General Solutions for Stresses and Displacements

The governing equations, Equations 2a and 2b, are coupled, linear partial differential equations with constant coefficients related to elastic constants of each fiber-reinforced lamina. Wang, *et al.* [13, 14] have shown that, by introducing proper forms for the Lekhnitskii complex-variable stress functions, the general solution for the problem can be expressed as

$$\sigma_x = \sum_{k=1}^{3} [C_k \mu_k^2 Z_k^{\delta} + C_{k+3} \mu_k^2 Z_k^{\delta}] + \sigma_{xo}, \tag{4a}$$

$$\sigma_y = \sum_{k=1}^{3} [C_k Z_k^{\delta} + C_{k+3} Z_k^{\delta}] + \sigma_{yo}, \tag{4b}$$

$$\tau_{yz} = \sum_{k=1}^{-3} [C_k \eta_k Z_k^{\delta} + C_{k+3} \eta_k Z_k^{\delta}] + \tau_{yzo}, \tag{4c}$$

$$\tau_{xz} = \sum_{k=1}^{3} [C_k \mu_k \eta_k Z_k^{\delta} + C_{k+3} \mu_k \eta_k Z_k^{\delta}] + \tau_{xzo}, \tag{4d}$$

$$\tau_{xy} = \sum_{k=1}^{-3} [C_k \mu_k Z_k^{\delta} + C_{k+3} \mu_k Z_k^{\delta}] + \tau_{xyo}, \tag{4e}$$

and

$$u = \sum_{k=1}^{3} [C_k p_k Z_k^{\delta+1} + C_{k+3} p_k \ Z_k^{\delta+1}]/(\delta+1) + u_o, \tag{5a}$$

$$v = \sum_{k=1}^{3} [C_k q_k Z_k^{\delta+1} + C_{k+3} q_k Z_k^{\delta+1}]/(\delta+1) + v_o, \tag{5b}$$

$$w = \sum_{k=1}^{3} [C_k \ t_k \ Z_k^{\delta+1} + C_{k+3} \ t_k \ Z_k^{\delta+1}]/(\delta+1) + w_o \ , \tag{5c}$$

where $Z_k = x + \mu_k y$, and μ_k and η_k are roots of the algebraic characteristic equations given in [13,16]. The p_k, q_k and t_k in Equations 5a-5c are related to lamina elastic constants and can also be found in References 13 and 16; σ_{io} and u_{jo} are known quantities determined from the remote loading condition for each individual case studied. The expression for σ_z can then be obtained as

$$\sigma_z = (e_3 - S_{3j} \ \sigma_j)/S_{33} \qquad (j = 1,2,4,5,6). \tag{5d}$$

4. DELAMINATION CRACK-TIP STRESS SINGULARITY

Consider a delamination between the kth and $(k + 1)$th plies in a composite laminate. Assuming that interlaminar crack surfaces are free from traction, one can immediately introduce the following stress boundary conditions:

$$\sigma_y^{(i)} = \tau_{yz}^{(i)} = \tau_{xy}^{(i)} = 0 \qquad (i = k,\ k + 1) \text{ on } \phi = \pm\ \pi. \tag{6a}$$

Continuity conditions for displacements and interlaminar stresses along the ply interface requires that

$$\{\sigma_y^{(k)}, \tau_{yz}^{(k)},\ \tau_{xy}^{(k)}\} = \{\sigma_y^{(k+1)}, \tau_{yz}^{(k+1)}, \tau_{xy}^{(k+1)}\}, \tag{6b}$$

$$\{u^{(k)}, v^{(k)}, w^{(k)}\} = \{u^{(k+1)}, v^{(k+1)}, w^{(k+1)}\}. \tag{6c}$$

Substituting Equations 4a–4e and 5a–5c into the above constraint conditions leads to twelve algebraic equations in $C_m^{(k)}$ and $C_m^{(k+1)}$. The existence of non-trival solutions for $C_m^{(k)}$ and $C_m^{(k+1)}$ requires that the determinant of coefficients of the twelve algebraic equations vanish, i.e.,

$$\|\Delta(\delta)\|_{12\times12} = 0. \tag{6d}$$

This gives a standard eigenvalue problem in which δ may be solved numerically from the transcendental equations, Equation 6d, involving material constants, S_{ij}, μ_k, η_k of plies adjacent to the delamination. The general form of the solution for δ can be written as

$$\delta_n = n \qquad \text{or} \qquad \delta_n = \left(n - \frac{1}{2}\right) \pm i\gamma \qquad (n = 0,1,2,\ .\ .\ .), \tag{7}$$

where γ is a real constant related to anisotropic elastic constants of fiber-reinforced laminae adjacent to the interlaminar crack only.

Owing to positive definiteness of strain energy in the elastic composite, the δ_n bounded by

$$-1 < \mathrm{Re}[\delta_n] < 0 \tag{8}$$

characterize the order of the delamination crack-tip stress singularity. For the delaminated $[\theta/-\theta/-\theta/\theta]$ graphite-epoxy composites with various fiber orientations, the eigenvalues δ_n, which satisfy the aforementioned constraint condition, are given in Table 1. The stress singularities for an interlaminar crack between two dissimilar, highly anisotropic laminae are observed to contain a pair of complex conjugates, $\delta_{1,2} = -0.5 \pm i\gamma$, and a constant, $\delta_3 = -0.5$. This situation is unique and different from that of an interface crack between two isotropic or orthotropic media [1–6] in the sense that δ_1, δ_2 and δ_3 always occur simultaneously in the present delamination problem. In the degenerated cases such as $\pm\theta = 0°$ and $90°$, the composite laminates become unidirectional. The delamination is located in an orthotropic material; thus, the classical inverse square-root singularity for crack-tip stresses is fully recovered.

From Equations 4, 5 and 7, it is clearly seen that the asymptotic stress field at a delamination crack tip possesses the well-known oscillatory singularity and the associated displacement field also exhibits oscillatory characteristics, leading to the controversial crack-surface overlapping or interpenetration, which is physically inadmissible. Similar phenomena have also been observed in studying an interface crack between dissimilar isotropic materials [1–9]. As first pointed out by England [5] and Malyshev, *et al.* [7] in examining the interface crack behavior in dissimilar isotropic solids, the crack surface overlapping is found to be confined to an extremely small region, and the interpenetration may not be of practical significance in the context of linear elastic fracture mechanics for an interface crack in a tensile field. Refined models to account for interface crack closure in dissimilar isotropic media have been reported in [10–12]. However, recent studies by Wang, *et al.* [14,15] have

Table 1. Dominant stress singularities for delamination in $[\theta/-\theta/-\theta/\theta]$ Graphite-epoxy composites.*

θ	$\delta_{1,2}$	δ_3
0°	−0.5	−0.5
15°	$-0.5 \pm 0.00642\,i$	−0.5
30°	$-0.5 \pm 0.02399\,i$	−0.5
45°	$-0.5 \pm 0.03434\,i$	−0.5
60°	$-0.5 \pm 0.02942\,i$	−0.5
75°	$-0.5 \pm 0.01579\,i$	−0.5
90°	−0.5	−0.5

*The composite ply properties given in [13,14] are used in the computation here.

shown that, in composite laminates with certain combinations of laminar elastic properties, ply orientations, and loading conditions, the delamination crack surface closure may become extremely large. Thus, under these circumstances, the current model and approach need to be modified to take into account the interlaminar crack closure and crack surface contact stresses.

5. STRESS INTENSITY FACTORS FOR DELAMINATION IN COMPOSITES

The eigenfunctions in Equations 4 and 5 are determined by imposing appropriate material and geometric symmetry conditions and remote boundary conditions. Hence, complete stresses and displacements, σ_i and u_i, in the composite laminate can be fully established. Neglecting the higher-order terms, the general asymptotic solution structure of the crack-tip stress field can be shown to have the following form:

$$\sigma_i = \sum_{j=1}^{3} \sum_{k=1}^{3} [f_{ijk}\, Z_k^{\delta_j} + g_{ijk}\, \overline{Z}_k^{\delta_j}] \qquad (i = 1,2,3, \ldots, 6), \tag{9a}$$

where Z and $\overline{Z}$ have their origin located at the delamination crack tip; f_{ijk} and g_{ijk} are related to material constants, laminate geometry, and boundary conditions, and δ_j are the eigenvalues bounded by Equation 8. For the convenience of further developments, the asymptotic stress field around the crack tip can be written as

$$\sigma_i = \sum_{j=1}^{3} \sigma_{ij}(x, y; \delta_j) + O(\text{non-singular, higher-order terms}), \tag{9b}$$

where σ_{ij} is the jth component of the singular stress σ_i corresponding to the eigenvalue δ_j which meets the constraint condition Equation 8.

In view of the stress solution structure given by Equations 9a and 9b, it is possible to define in the context of mechanics of fracture the delamination stress intensity factors for the composite in a manner analogous to that given in References 4–6 by considering the interlaminar stresses, σ_2, σ_4 and σ_6 (i.e., σ_y, τ_{yz}, and τ_{xy}), along the ply interface ahead of a delamination crack as

$$K_I = \lim_{x \to 0^+} \sum_{j=1}^{3} \sqrt{2\pi}\, x^{-\delta_j}\, \sigma_{2j}(x, 0; \delta_j), \tag{10a}$$

$$K_{II} = \lim_{x \to 0^+} \sum_{j=1}^{3} \sqrt{2\pi}\, x^{-\delta_j}\, \sigma_{6j}(x, 0; \delta_j), \tag{10b}$$

$$K_{III} = \lim_{x \to 0^+} \sum_{j=1}^{3} \sqrt{2\pi}\, x^{-\delta_j}\, \sigma_{4j}(x, 0; \delta_j), \tag{10c}$$

in which the origin of the Cartesian coordinates has been transferred to the crack tip.

It is noted that the K_I, K_{II} and K_{III} defined in Equations 10a–10c for a delamination crack between dissimilar anisotropic composites are different from those for a crack in a homogeneous solid. Thus, the stress intensity factors may not carry the conventional physical interpretation as in the cohesive (or homogeneous) fracture. While the overall stress intensity factor [3–7] may be used to depict the maximum amplitude of crack-tip stress and to correlate crack extension, accurate description of the asymptotic stress field at the composite delamination crack tip requires detailed knowledge of individual stress intensity factors. Because of the complexities involved in the problem, numerical methods are generally required to obtain this information.

6. STRAIN ENERGY RELEASE RATE FOR DELAMINATION

While the stress intensity factors K_I, K_{II} and K_{III} describe the details of an interlaminar crack-tip field, the strain energy release rate G is also of significant interest, since this is a quantity physically measurable in experiments and mathematically well defined. The fracture energy release rate, G, in a delaminated composite may be evaluated by using Irwin's virtual crack extension concept [17] as

$$\begin{aligned} G =& \quad G_I + G_{II} + G_{III} \\ =& \lim_{\delta\beta\to 0} \frac{1}{2\delta\beta}\int_0^{\delta\beta} \{\sigma_y(r,0)[v^{(k)}(\delta\beta - r, \pi) - v^{(k+1)}(\delta\beta - r, -\pi)] \\ & + \tau_{xy}(r,0)[u^{(k)}(\delta\beta - r, \pi) - u^{(k+1)}(\delta\beta - r, -\pi)] \\ & + \tau_{yz}(r,0)[w^{(k)}(\delta\beta - r, \pi) - w^{(k+1)}(\delta\beta - r, -\pi)]\}\,dr, \end{aligned} \tag{11}$$

where polar coordinates (r, ϕ) are used for the convenience of computation, and $\delta\beta$ is the length of virtual crack extension. The interlaminar stresses, σ_y, τ_{xy}, and τ_{yz}, in Equation 11 can be obtained from the crack-tip stress field equations such as Equations 4a–4e. The corresponding displacements are also those of the asymptotic field equations obtained in the previous section, i.e., Equations 5a–5c.

It is noted here that, even though the near-field stresses possess an oscillatory singularity and $K_i (i = I, II, III)$ may not have the usual significance attached as in the cohesive (homogeneous) crack case, the energy release rates, G_i and G, are well defined quantities in the problem, and should be of practical importance in studying the composite delamination. The G and its components G_I, G_{II} and G_{III} may be evaluated theoretically and experimentally (in conjunction with proper analyses) to provide basic measures for delamination problems.

7. NUMERICAL EXAMPLES AND DISCUSSION

For the purpose of illustrating the fundamental nature of the delamination crack behavior in composite materials, graphite-epoxy laminates with symmetric $[\theta/-\theta/-\theta/\theta]$ fiber orientations containing delamination cracks along the θ and $-\theta$ ply interface are studied here. The composite laminate is subjected to a uniform axial extension and has a geometry shown in Figure 1 with a width-to-thickness ratio $2b/2W$ and uniform ply thicknesses h_i. Delamination cracks of length a are assumed to emanate from edges of the composite. Ply properties typical of high-modulus unidirectional graphite-epoxy composite given in [13,14] are used in the numerical computation.

7.1 Influence of Fiber Orientation on Delamination Crack Behavior

Consider the $[\theta/-\theta/-\theta/\theta]$ graphite-epoxy composites with various fiber orientations subjected to uniform axial strain, $\varepsilon_z = e$. For illustration, we choose the composite with an aspect ratio $2b/(2h)$ equal to 8, ply thickness $h_1 = h_2 = 1$ in., and delamination length $a = 1$ in. The mixed-mode K_I, K_{II} and K_{III} are determined and given in Figure 2 for various θ's. It is observed that even though the composite laminate is under the simplest loading condition, the delamination crack-tip response is very complicated indeed due to the complex interlaminar stress transfer, the nonhomogeneity of the composite, the anisotropy of ply properties, and the unusual delamination con-

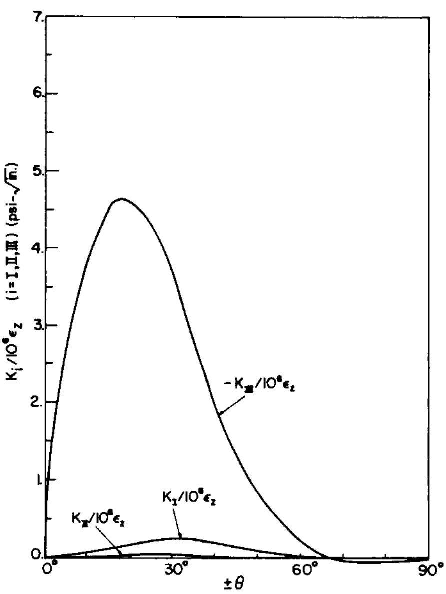

Figure 2. Delamination stress intensity factors, K_i (i = I, II, III), in angle-ply $[\theta/-\theta/-\theta/\theta]$ graphite-epoxy composite under uniform axial extension ε_z ($h_1 = h_2 = 1$ in., $b = 8$ in., $a = 1$ in.).

figuration with respect to the loading direction. The tearing-mode stress intensity factor K_{III} associated with the interlaminar shear τ_{yz} is about one or two orders of magnitude higher than K_I and K_{II} in general in the laminates studied. The opening-mode stress intensity K_I is also very appreciable due to the significant interlaminar normal stress σ_y. The simultaneous presence of K_I, K_{II} and K_{III} in a composite delamination problem is unique to fiber-reinforced laminates, not observable in fracture problems of bonded dissimilar isotropic media in general. The total strain energy release rates, G, for the $[\theta/-\theta/-\theta/\theta]$ graphite-epoxy composite laminates with the aforementioned delamination are shown in Figure 3. The results reveal that the energy release rate, i.e., the driving force for delamination fracture, is highest for a delaminated $[\pm\theta]_s$ composite with $\theta \cong 18°$, and has rather small values for $\theta \geqslant 45°$, indicating that the composite with $\theta \geqslant 45°$ may be more prone to intralaminar cracking than delamination. Obviously, from the present results, the delamination behavior is observed to be inherently three-dimensional in nature. Thus, for composites with more general lamination, crack geometry, and loading conditions, fully three-dimensional fracture analyses are essential for obtaining complete information.

7.2 Effect of Ply Thickness on Delamination

The effect of laminate geometric variables on the delamination behavior is best illustrated by examining the change of K_I, K_{II} and K_{III} with ply thickness in a $[45°/-45°/-45°/45°]$ graphite-epoxy composite with $h_1 + h_2 = W = 2$ in. Given the delamination crack length ($a = 1$ in.), laminate width ($2b =$

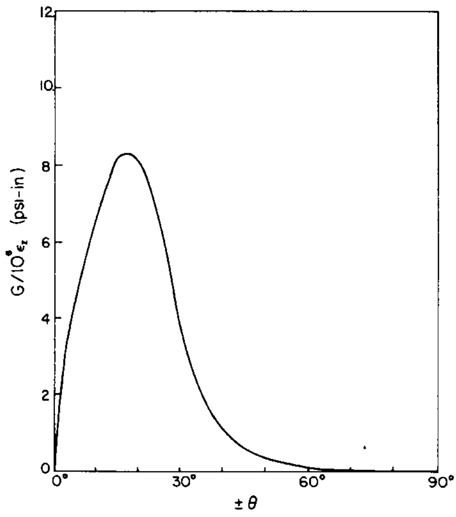

Figure 3. Energy release rate G for edge delamination in angle-ply $[\theta/-\theta/-\theta/\theta]$ graphite-epoxy composite under uniform axial extension ε_z ($h_1 = h_2 = 1$ in., $b = 8$ in., $a = 1$ in.).

4 in.), and the loading condition as before, the delamination stress intensity factors for various h_1/h_2 cases are obtained in Figure 4. The crack-tip tearing and opening stresses, τ_{yz} and σ_y (or K_{III} and K_I), have maximum intensifications, as the ply thicknesses h_i become identical, i.e., $h_1/h_2 \cong 1$. The K_{II}, however, reaches a minimum due to the reduction in τ_{xy} in this case. It should be noted here that the K_I, K_{II} and K_{III} depend on material constants of all plies as well as the overall laminate geometry. Therefore, values of K_i are related not only to local crack properties but also to global lamination and geometric variables as well as the loading condition.

The delamination strain energy release rate is also a function of ply thickness. In the [45°/−45°/−45°/45°] graphite-epoxy composite with a geometry given before, the change of G with h_1/h_2 is shown in Figure 5, where maximum driving force occurs at $h_1 = h_2$, indicating the criticality of relative ply thickness for delamination fracture in composites.

7.3 Interlaminar Crack Length vs. Delamination Energy Release Rate

To illustrate the basic nature of delamination extension in laminated composites, strain energy release rates in the [45°/−45°/−45°/45°] graphite-epoxy with various interlaminar crack lengths are presented. Effects of laminate width on delamination crack extension are also investigated. The change of total strain energy release rate G with delamination length a is given in Figure 6 to illustrate fundamental characteristics of delamination fracture. For the composite laminates with various $2b/2h$'s, the G is observed to change with delamination length in a unique manner. The energy release rate (or crack-extension driving force) reaches a maximum or a relative constant at a length a^* approximately equal to one or two ply thicknesses in the com-

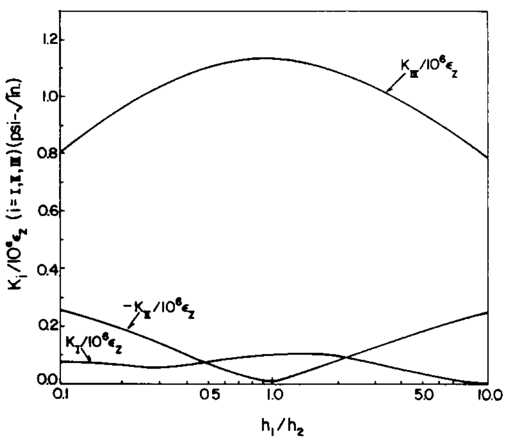

Figure 4. Stress intensity factors, K_I, K_{II} and K_{III}, of an edge delamination crack in [45°/−45°/−45°/45°] graphite-epoxy composite with various ply-thickness ratios h_1/h_2 (a = 1 in., W = 2 in., 2b = 4 in.).

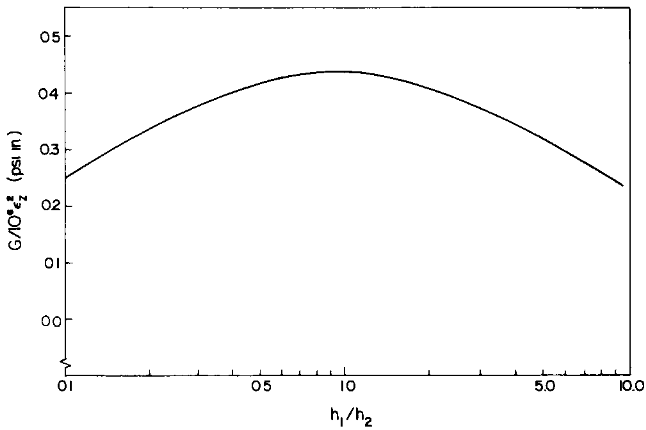

***Figure 5.** Effect of ply thickness h_1/h_2 on strain energy release rate G in [45°/−45°/−45°/45°] graphite-epoxy composite (a = 1 in., 2b = 4 in., W = 2 in.).*

posites studied, depending upon the $(2b/2h)$ ratio. As the delamination exceeds this characteristic dimension a^*, the G either decreases monotonically or remains constant.

From a fracture mechanics perspective, several important features regarding the basic nature of composite delamination are revealed. Assuming that the material resistance to delamination growth remains constant (i.e., the failure criterion, G_c = constant, is used), one may find that a critical dimension of delamination length, a*, associated with the maximum G for each composite laminate may exist (e.g., $a^* \cong 2h$ for the case $b/h = 8$); the word

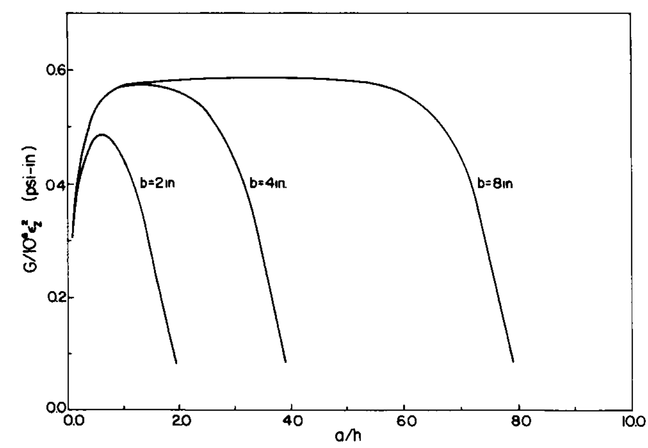

***Figure 6.** Delamination strain energy release rate G in [45°/−45°/−45°/45°] graphite-epoxy composite with various delamination crack lengths a and laminate widths b ($h_1 = h_2 = 1$ in.).*

"critical" refers to the one that experiences a stable crack extension at the lowest load. It also indicates that any inherent interlaminar flaw a_o in the composite, which is less than a^*, will experience initial unstable growth as the load or G reaches the critical level, and is anticipated to be arrested at a later stage. Any initial delamination greater than a^* will experience a stable growth under a monotonically rising load. The phenomena explained by the $G - a$ curve have been noted by several researchers conducting experimental and analytical studies on the delamination fracture. The a^* may be an important quantity in the interlaminar toughness measurement and life prediction for composite materials and structures subjected to static and cyclic loading.

8. CONCLUDING REMARKS

Based on the information obtained in the paper, the following remarks may be suggested:

1. Stress singularities for a delamination crack between dissimilar, highly anisotropic laminae are obtainable by using an eigenfunction analysis. The order of delamination crack-tip stress singularity is different from that of an interface crack between dissimilar isotropic or orthotropic media by the simultaneous presence of three distinct characteristic eigenvalues, $-0.5 + i\gamma$, $-0.5 - i\gamma$, and -0.5.
2. The fracture mechanics concept may be extended to delamination problems in anisotropic composite laminates by properly defining the interlaminar crack-tip intensity factors and strain energy release rates. For edge-delaminated angle-ply composite laminates, K_I, K_{II} and K_{III} always occur simultaneously with K_{III} being one or two orders of magnitude higher than the other two.
3. The crack-extension driving force or strain energy release rate for delamination in composite laminates can be determined using Irwin's virtual crack extension concept. Delamination stability in composite laminates may be assessed for any inherent interlaminar flaw relative to the critical delamination size a^* obtained in the $G - a$ diagram.
4. Fiber orientation, ply thickness, stacking sequence and other related lamination variables are found to influence the delamination crack behavior (in terms of K_I, K_{II}, K_{III} and G) significantly.

9. ACKNOWLEDGEMENT

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